

$$\begin{aligned}
& \text{Diagram 1} + \text{Diagram 2} + \text{Diagram 3} + \text{Diagram 4} = \text{disc}_{p_1^2} \text{disc}_{p_2^2} \int \frac{d^4 k}{(2\pi)^4} \frac{\mathcal{N}_{\text{box}} \delta_+(k)}{p_1^2 p_2^2 (p_2 - k)^2 (k - p_1)^2}
\end{aligned}$$